1 ASTROME
FOR TELECOM INFRASTRUCTURE

How can we actually leverage 5G technology to improve connectivity, even in rural areas? Let's find out.

The Gigamesh Pathway gives a 10km range with 2Gbps speed per link, with up to 4 links. It is suitable for 5G backhaul, rural broadband, and enterprise and defense connectivity

The Gigamesh 5G Edge is the next version of the Gigamesh Pathway. It will not only scale the markets that it addresses, but also its capacity per link

Astrome is a deep-tech startup from Bengaluru that uses its proprietary technology to accelerate the construction of 5G and rural telecommunication infrastructure. Founded by Dr Prasad Bhat and Dr Neha Satak in 2015, Astrome aims to help India and the world connect better. Apart from Bengaluru, the company has a presence in the USA as well.

During this pandemic, the world realised that connectivity is nothing less than a necessity—not just in cities but also in rural areas and in the defense sector. "If you look at the telecom infrastructure around you, there is great infrastructure whenever you have fibre. But when you move away from fibre, the kind of connectivity you get is not really adequate," says Dr Satak. "This problem is only going to intensify when you deploy 5G."

The Market Verticals of Astrome

- Defense Communication
- Rural Broadband
- 4G/5G Telecom Networks
- Enterprise and Private Networks

More about Gigamesh

Scalability in terms of range. It can work in urban areas where you need a shorter range and a higher capacity, and also in rural areas where you need a very large range of more than 10km.

Scalability in terms of capacity. It can deliver from 1Gbps to 100Gbps speeds in the future versions.

Completely software-driven. Since the system is controlled and managed through software, it reduces the overall cost.

The only solution is to have wireless equipment that can give gigabit speeds or higher, but at a good price point. To accomplish all of that, we have to develop really cool, cost-effective wireless technologies that can be deployed fast. These technologies need to complement fibre optics for us to get connected better and faster.

To solve these problems, Astrome has developed Gigamesh. This product has won several awards, one of them being "The most promising connectivity solution" from the International Telecommunication Union. It is a patented technology that happens to be the world's first multibeam E-band radio. This means that it works at very high frequencies. It can do the job of multiple fibre strands to connect one tower, which is connected with fibre to multiple towers that are not connected with fibre.